

LEARNING COLOR IMAGE EXPANSION FILTERS

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ABSTRACT

Image expansion by linear filtering is attractive and widely used because of its simplicity and efficiency, and many interpolation methods fall in this category. In this study, we model filtering as linear regression from low- to high-resolution color image patches, and propose a learning-based design method of image expansion filters based on sparse Bayesian estimation. Sparseness is imposed on the filter coefficients to obtain compact supports. Image expansion is formulated as the problem of finding the predictive mean of a high-resolution patch given a low-resolution patch to expand. Since an exact evaluation of the predictive distribution is difficult, variational methods are employed to derive an efficient algorithm. Experiments on test data show that good generalization performance is obtained based on sparse filters and that color modeling improves the expansion quality.

Index Terms—Image expansion, interpolation, resolution synthesis, sparse Bayesian learning, variational inference

1. INTRODUCTION

Letting r be an integer magnification factor, we state the task of image expansion as follows:

from a given $M \times N$ image ξ , estimate an $rM \times rN$ expanded image $\hat{\xi}$.

The interpolator expands the image by replacing each one pixel in the given image by an $r \times r$ high-resolution image patch. Since it is impossible to estimate r^2 pixel values from only one pixel value, we use the low-resolution patch surrounding the pixel to be replaced (Fig. 1). This local interpolation is repeated for every pixel in the given image, and the expanded image is constructed by tessellating the high-resolution patches.

Classical image interpolators are designed according to some top-down assumptions, e.g., the image is piecewise polynomials that are smooth at the knots; but these assumptions are not guaranteed to work best for natural images. For example, the bilinear interpolator uses four low-resolution pixels to estimate the corresponding high-resolution pixel, and it is well known that it produces blurred high-resolution images (typically mended by a post-sharpening heuristic).

Alternatively, filter design strategies based on learning from real image data have shown to exhibit superior performance compared to classical image interpolators [1–3], where filters are trained using datasets consisting of high-resolution images and low-resolution images generated by synthetically smoothing and subsampling the high-resolution images. Triggs’ approach [1] is to learn filter coefficients by minimizing interpolation errors in a maximum likelihood manner. Sparse Bayesian estimation [4] was employed by Kanemura et al. [2], whose learning algorithm can reveal compact supports that

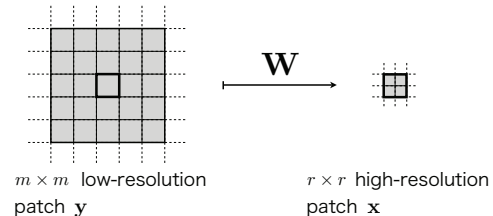


Fig. 1. $m \times m$ low-resolution patch is used to estimate $r \times r$ high-resolution patch (the case of $m = 5, r = 2$ is shown). The thick squares indicate the same position.

are relevant to image expansion. A related approach is resolution synthesis by Atkins [3], where a mixture of linear interpolators is trained through the expectation-maximization (EM) algorithm.

Since it is essential for realistic applications to handle color images, in this study we consider all the three color channels both in the YIQ and RGB domains. The emphasis of Triggs [1] is low-level vision and he considered only grayscale pixel values. The interpolator of Kanemura et al. [2] expands color images by applying learned filters only to the luminance component in the YIQ domain, and the other two channels are expanded by the cubic interpolator. Atkins [3] used channel-wise interpolators that are trained independently for each of the RGB channels.

The sinc function is the idealized interpolator, but its use is impractical partly because of its infinite support [5]. Simple truncation of the sinc function results in deteriorated interpolation performance. Then, some interpolators are designed to approximate the sinc function. Lehmann et al. [5] catalogues a number of classical (i.e., non-learning-based) interpolation kernels with various shapes and finite support sizes, and compares their performance on grayscale image manipulations.

2. SPARSE BAYESIAN MODEL FOR IMAGE EXPANSION FILTERS

In this section, we develop a synthetic model of image expansion by defining the joint probability distribution of random variables.

2.1. Synthetic modeling

Let $\mathbf{y}$ be the Q -dimensional vector of a low-resolution patch, and let $\mathbf{x}$ be the D -dimensional vector of a high-resolution patch. The vectors $\mathbf{y}$ and $\mathbf{x}$ are made by forming pixel values in lexicographic order (and hence $Q = m^2, D = r^2$). We assume that the high-resolution patch $\mathbf{x}$ is synthesized from the corresponding low-resolution patch $\mathbf{y}$ by the linear operation:

$$\mathbf{x} = \mathbf{W}\mathbf{y} + \boldsymbol{\varepsilon}, \quad (1)$$

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where $\mathbf{W}$ is a $D \times Q$ matrix and ϵ is Gaussian noise with precision (inverse variance) β . This equation defines the likelihood in our model:

$$p(\mathbf{x}|\mathbf{y}, \mathbf{W}, \beta) = \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{y}, \beta^{-1}\mathbf{I}_D), \quad (2)$$

where $\mathcal{N}(\cdot)$ is a Gaussian distribution¹ and $\mathbf{I}_D$ is a D -dimensional identity matrix.

If we let $\mathbf{w}_d^T$ be the d th row of $\mathbf{W}$ (that is, $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_D]^T$), then we have $x_d = \mathbf{w}_d^T \mathbf{y} + \epsilon_d$. Therefore, each row of $\mathbf{W}$ is the filtering kernel for expanding the low-resolution image.

We consider the following three schemes for handling color images.

- Only the luminance in YIQ [2]:

$$\mathbf{x}^Y = \mathbf{W}^Y \mathbf{y}^Y + \epsilon^Y. \quad (3)$$

The other IQ channels are expanded by a classical technique.

- RGB:

$$\mathbf{x}^c = \mathbf{W}^c \mathbf{y}^c + \epsilon^c \quad \text{for } c \in \{\mathbf{R}, \mathbf{G}, \mathbf{B}\}. \quad (4)$$

- YIQ:

$$\mathbf{x}^c = \mathbf{W}^c \mathbf{y}^c + \epsilon^c \quad \text{for } c \in \{\mathbf{Y}, \mathbf{I}, \mathbf{Q}\}. \quad (5)$$

2.2. Prior probabilities

In a Bayesian treatment, $\mathbf{W}$ and β are considered random variables, and prior distributions with parameter $\mathbf{A} = [\alpha_{dq}]$ are placed on them. Our priors are:

$$p(\mathbf{W}|\mathbf{A}) = \prod_{d=1}^D \prod_{q=1}^Q \mathcal{N}(w_{dq}|0, \alpha_{dq}^{-1}), \quad (6)$$

$$p(\beta) = \text{Gam}(\beta|a_{\beta 0}, b_{\beta 0}), \quad (7)$$

where $\text{Gam}(\cdot)$ is a gamma distribution² and $a_{\beta 0}, b_{\beta 0}$ are hyperparameters. On $\mathbf{A}$, we further put a prior distribution with hyperparameters $a_{\alpha 0}, b_{\alpha 0}$

$$p(\mathbf{A}) = \prod_{d=1}^D \prod_{q=1}^Q \text{Gam}(\alpha_{dq}|a_{\alpha 0}, b_{\alpha 0}). \quad (8)$$

The hyperparameters subscripted with 0 are determined in advance. For $p(\beta)$, we take the noninformative limit $a_{\beta 0} \rightarrow 0, b_{\beta 0} \rightarrow 0$. For $p(\alpha)$, we take $b_{\alpha 0} \rightarrow 0$ but $a_{\alpha 0} \neq 0$ to enhance the sparsity of the filter coefficients, and thus we call $a_{\alpha 0}$ the *sparsity hyperparameter*, which is the only hyperparameter controlled manually.

The prior for $\mathbf{W}$ in our model, (6), resembles that in sparse Bayesian estimation [4], which leads the mechanism called automatic relevance determination (ARD). ARD works effectively to select only elements that satisfy a certain condition, resulting in a sparse solution. If α_{dq} is small, the prior belief gets stronger and the probability distribution of w_{dq} becomes concentrated around the mean value 0. If α_{dq} goes infinity, the probability approaches the delta function and the filter coefficient w_{dq} will always be 0, pruning the dq element away. It is known that α_{dq} diverges to infinity when the dq element is not “relevant” [4], and indeed it has been shown that compact supports that exhibit superior performance to the classical interpolators can be automatically selected using ARD [2].

¹ $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = |\mathbf{2}\pi\boldsymbol{\Sigma}|^{-1/2} \exp\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\}$.

² $\text{Gam}(\tau|a, b) = \Gamma(a)^{-1} b^a \tau^{a-1} \exp\{-b\tau\}$, defined for $\tau \geq 0$.

3. INFERENCE FOR EXPANDING IMAGES

In this section we describe Bayesian inference for estimating a high-resolution patch given a low-resolution patch based on our probability model.

Let $\mathcal{D} = \{(\mathbf{x}_n, \mathbf{y}_n) \mid n = 1, \dots, N\}$ be the dataset from which the filter is learned. For notational simplicity we let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]$ and $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_N]$. From the probability model defined in the previous section, we have the joint distribution

$$p(\mathbf{A}, \mathbf{W}, \beta, \mathbf{X}|\mathbf{Y}) = p(\mathbf{A})p(\mathbf{W}|\mathbf{A})p(\beta) \prod_{n=1}^N p(\mathbf{x}_n|\mathbf{y}_n, \mathbf{W}, \beta), \quad (9)$$

which is a product of (2), (6)–(8). Inference regarding the variables other than the high-resolution patch is embraced by the posterior distribution

$$p(\mathbf{A}, \mathbf{W}, \beta|\mathcal{D}) = \frac{p(\mathbf{A}, \mathbf{W}, \beta, \mathbf{X}|\mathbf{Y})}{\int p(\mathbf{A}, \mathbf{W}, \beta, \mathbf{X}|\mathbf{Y}) d\mathbf{A}d\mathbf{W}d\beta}, \quad (10)$$

which is derived from Bayes’ theorem.

Given a new low-resolution patch $\mathbf{y}$ to expand, Bayesian inference utilizes the predictive distribution of the corresponding high-resolution patch $\mathbf{x}$, which is calculated from the likelihood (2) and the posterior (10) as

$$p(\mathbf{x}|\mathbf{y}, \mathcal{D}) = \int p(\mathbf{x}|\mathbf{y}, \mathbf{W}, \beta) p(\mathbf{A}, \mathbf{W}, \beta|\mathcal{D}) d\mathbf{A}d\mathbf{W}d\beta. \quad (11)$$

Then the estimate for the high-resolution image is the predictive mean

$$\mathbb{E}(\mathbf{x}) = \int \mathbf{x} p(\mathbf{x}|\mathbf{y}, \mathcal{D}) d\mathbf{x}. \quad (12)$$

Unfortunately, it is difficult to perform the required integrations analytically. Hence we utilize variational methods to have tractable computations.

4. VARIATIONAL APPROXIMATION

In this section, we derive an efficient algorithm for approximate Bayesian inference based on variational methods [6]. Let the random variables denoted by $\boldsymbol{\eta} = (\mathbf{A}, \mathbf{W}, \beta)$. Then the posterior distribution is denoted by $p(\boldsymbol{\eta}|\mathcal{D})$. We approximate $p(\boldsymbol{\eta}|\mathcal{D})$ by another distribution $q(\boldsymbol{\eta})$, which we call a trial distribution and force it to have a tractable form. We optimize q to be the one that minimizes Kullback-Leibler (KL) divergence to the true p :

$$q^*(\boldsymbol{\eta}) = \arg \min_q D_{\text{KL}}(q(\boldsymbol{\eta})||p(\boldsymbol{\eta}|\mathcal{D})). \quad (13)$$

KL divergence is defined as

$$D_{\text{KL}}(q(\boldsymbol{\eta})||p(\boldsymbol{\eta}|\mathcal{D})) = -\left\langle \ln \frac{p(\boldsymbol{\eta}|\mathcal{D})}{q(\boldsymbol{\eta})} \right\rangle_{\boldsymbol{\eta}}, \quad (14)$$

where $\langle \cdot \rangle_{\boldsymbol{\eta}}$ is an expectation operator with respect to $q(\boldsymbol{\eta})$.

If we allowed q to be free from any constraints, the optimal $q^*(\boldsymbol{\eta})$ would be the trivial solution $p(\boldsymbol{\eta}|\mathcal{D})$, which is difficult to compute. If we impose factorization assumption on q , i.e., $q(\boldsymbol{\eta}) = \prod_i p(\boldsymbol{\eta}_i)$, the optimal factor $q^*(\boldsymbol{\eta}_i)$ given the other factors $q(\boldsymbol{\eta}_{\setminus i}) = \prod_{j \neq i} q(\boldsymbol{\eta}_j)$ is expressed as [6]

$$\ln q^*(\boldsymbol{\eta}_i) = \langle \ln p(\boldsymbol{\eta}, \mathbf{X}|\mathbf{Y}) \rangle_{\boldsymbol{\eta}_{\setminus i}} + \text{const.}, \quad (15)$$

which can be evaluated analytically in our model. We present the optimal factor-distributions $q^*(\mathbf{A})$, $q(\mathbf{W})$, and $q^*(\beta)$ in the following subsection. Since they depend on each other, we have to iterate computing them in order to find a joint optimum.

4.1. Optimal factors

The optimal factor for $\mathbf{A}$ is found to be an element-wise independent distribution

$$q^*(\mathbf{A}) = \prod_{d=1}^D \prod_{q=1}^Q \text{Gam}(\alpha_{dq} | a_\alpha, b_{\alpha dq}), \quad (16)$$

where

$$a_\alpha = a_{\alpha 0} + \frac{1}{2}, \quad b_{\alpha dq} = b_{\alpha 0} + \frac{\langle w_{dq}^2 \rangle}{2}. \quad (17)$$

The optimal factor-distribution of $\mathbf{W}$ is a product of row-wise Gaussian distributions

$$q^*(\mathbf{W}) = \prod_{d=1}^D \mathcal{N}(\mathbf{w}_d | \boldsymbol{\mu}_d, \boldsymbol{\Sigma}_d), \quad (18)$$

where

$$\boldsymbol{\Sigma}_d = \left(\text{diag}(\langle \alpha_{d1} \rangle, \dots, \langle \alpha_{dQ} \rangle) + \langle \beta \rangle \sum_{n=1}^N \mathbf{y}_n \mathbf{y}_n^T \right)^{-1}, \quad (19)$$

$$\boldsymbol{\mu}_d = \langle \beta \rangle \boldsymbol{\Sigma}_d \sum_{n=1}^N x_{nd} \mathbf{y}_n. \quad (20)$$

We put $\mathbf{M} = [\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_D]^T$ for denoting the expected value of the matrix $\mathbf{W}$.

Finally, the factor-distribution of β that minimizes KL divergence is given by

$$q^*(\beta) = \text{Gam}(\beta | a_\beta, b_\beta), \quad (21)$$

where

$$a_\beta = a_{\beta 0} + \frac{ND}{2}, \quad b_\beta = b_{\beta 0} + \frac{1}{2} \sum_{n=1}^N \langle \|\mathbf{x}_n - \mathbf{W} \mathbf{y}_n\|^2 \rangle. \quad (22)$$

The expectations in the above equations are evaluated as

$$\langle \|\mathbf{x}_n - \mathbf{W} \mathbf{y}_n\|^2 \rangle = \|\mathbf{x}_n - \mathbf{M} \mathbf{y}_n\|^2 + \text{tr} \left(\mathbf{y}_n \mathbf{y}_n^T \sum_{d=1}^D \boldsymbol{\Sigma}_d \right), \quad (23)$$

$$\langle w_{dq}^2 \rangle = m_{dq}^2 + [\boldsymbol{\Sigma}_d]_{qq}, \quad (24)$$

$$\langle \alpha_{dq} \rangle = a_\alpha / b_{\alpha dq}, \quad (25)$$

$$\langle \beta \rangle = a_\beta / b_\beta. \quad (26)$$

4.2. Approximate inference for expanding images

The objective of computing (12) is resolved by calculating the expectation with respect to the trial distribution:

$$\mathbb{E}(\mathbf{x}) \approx \langle \mathbf{x} \rangle = \langle \mathbf{W} \mathbf{y} + \boldsymbol{\epsilon} \rangle = \langle \mathbf{W} \rangle \mathbf{y} = \mathbf{M} \mathbf{y}. \quad (27)$$

Although it appeared that the Bayesian estimate given by (12) is extremely complicated, the resulting image-expansion equation (27) is just a linear filtering operation with the expectation of $\mathbf{W}$ as the filtering kernel.

Input: High- and low-resolution patch pairs $\mathcal{D} = (\mathbf{X}, \mathbf{Z})$ and sparsity hyperparameter $a_{\alpha 0}$

Output: Filter parameter $\mathbf{M}_W$

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1: repeat
2:   Update  $q(\mathbf{A})$  by computing  $a_\alpha$  and  $b_{\alpha dq}$ 
3:   if  $\langle \alpha_{dq} \rangle > e^{20}$  then
4:      $\langle \alpha_{dq} \rangle \leftarrow \infty$ 
5:   end if
6:   Update  $q(\mathbf{W})$  by computing  $\boldsymbol{\Sigma}_d$  and  $\boldsymbol{\mu}_d$ 
7:   Update  $q(\beta)$  by computing  $a_\beta$  and  $b_\beta$ 
8: until  $\Delta < 10^{-6}$ 

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Fig. 2. Algorithm for learning parameters for image expansion.

4.3. Training algorithm

To find the kernel parameter $\mathbf{M}$, or to train the expander, we use the algorithm shown in Fig. 2. To accelerate convergence, $\langle \alpha_{dq} \rangle$ is thresholded if it is sufficiently large. The criterion for convergence assessment is the relative change of the Frobenius norm of $\mathbf{M}$:

$$\Delta = \|\mathbf{M}' - \mathbf{M}\|_F / \|\mathbf{M}'\|_F, \quad (28)$$

where $\mathbf{M}'$ is the matrix at the previous iteration step.

By taking the symmetry of natural image patches into account, we wish to impose symmetry on the filters for different d . Since the kernel $\mathbf{w}_d$ that is used for estimating x_d is regularized by α_{dq} ($q = 1, \dots, Q$), we constrain α_{dq} to be horizontally and vertically symmetric over different d .

5. EXPERIMENTAL RESULTS

We conduct experiments to see the effect of color modeling on learning image expansion filters. Images used for training are #4.1.[01–08] in the USC-SIPI image database [7], and test images are #4.2.[01–07] in [7]. They are stored as integer RGB values ranging from 0 to 255. The pixel values are converted into double precision floating point within range $[0, 1]$ and all subsequent computations are performed in this domain. The expanding factor and the size of low-resolution patches are chosen to be $r = 2$ and $m = 11$, respectively. High-resolution patches are prepared by cutting images into non-overlapping pieces of size $r \times r$. To make low-resolution patches, the high-resolution images are blurred by the cubic kernel for anti-aliasing and subsampled by a factor of r and then overlapping patches of size $m \times m$ are extracted. For the training dataset, low-resolution patches stemming from the boundaries are discarded, and the corresponding high-resolution patches are not used. For the test dataset, the low-resolution image is extended by pixel replication.

The quality of an expanded image $\hat{\xi}$ is measured by peak signal-to-noise ratio (PSNR), which is defined using the true high-resolution image ξ by

$$\text{PSNR}(\xi, \hat{\xi}) = 10 \log_{10} \frac{1^2}{\text{MSE}(\xi, \hat{\xi})}, \quad (29)$$

where $\text{MSE}(\xi, \hat{\xi})$ is the mean squared error between ξ and $\hat{\xi}$ calculated in the RGB domain.

Using the eight training images, we trained image expansion filters of the three color schemes with the sparsity hyperparameter $a_{\alpha 0}$ controlled from 5 to 210. Each trained expander's mean PSNRs for the seven test images are plotted in Fig. 3. The YIQ filter has higher PSNRs with smaller supports compared to the RGB filter. Both RGB and YIQ expanders have better performance than the Y + cubic expander.

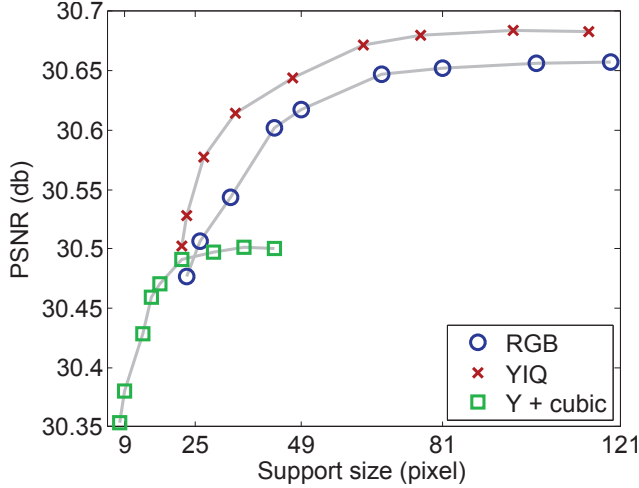


Fig. 3. Performances of image expansion filters.

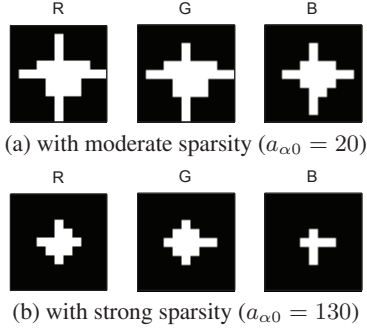


Fig. 4. Learned support of RGB filters.

Fig. 4 shows the supports of the trained RGB filter with moderate sparsity of $a_{\alpha 0} = 20$ and strong sparsity of $a_{\alpha 0} = 130$. As the sparsity gets stronger, the supports become smaller. The G channel appears to have the biggest support, which suggests that, in natural images, the green component has wider spatial correlation. Fig. 5 shows the supports of the learned filters by the YIQ scheme with $a_{\alpha 0} = 20$ and $a_{\alpha 0} = 130$. Again, the supports become smaller as the sparsity hyperparameter increases, and it seems the Y channel has wider supports than the other channels.

Fig. 6 displays images expanded by the cubic filter and the YIQ filter with $a_{\alpha 0} = 20$. The learning-based YIQ filter has approximately 2 dB better PSNR than the cubic method.

6. CONCLUSION

We have extended the sparse Bayesian image expansion filter learning algorithm to YIQ and RGB color domains. It has been shown that considering all the three color channels improves the expansion performance compared the single-channel expander and that the YIQ expanders are superior to the RGB expanders.

7. REFERENCES

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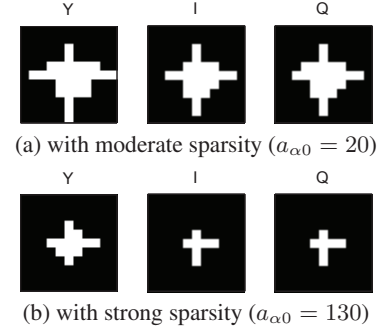


Fig. 5. Learned support of YIQ filters.

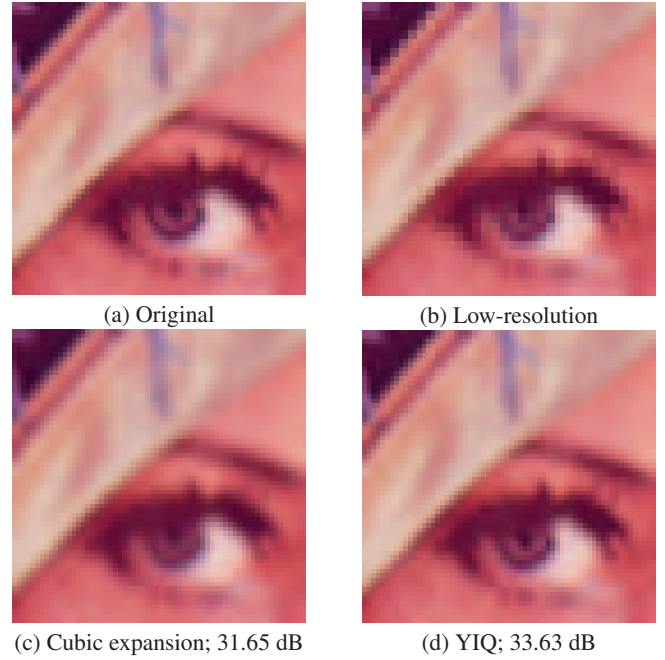


Fig. 6. The original image (a) is shrunk to be the low-resolution image (b), which is expended by the cubic filter (c) and the YIQ filter learned with $a_{\alpha 0} = 20$ (d).

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